Agents and Tool Use

Connecting a language model to the rest of the world

A question a language model cannot answer

"Will it rain in Yerevan today?"

This is not a hard question. It is an **impossible** one, and for a reason no amount of training fixes: the answer did not exist when the model was trained, and it will be different tomorrow.

A language model produces **text**. It has no clock, no network, no filesystem. Whatever it says about today's weather, it made up.

Give it a way to look

Now suppose the model can ask for one thing to be run on its behalf:

```
get_weather(city="Yerevan")
→ {"temp_c": 20, "condition": "clear", "rain_prob": 0.05}
```

"No rain expected in Yerevan today - clear, around 20 degrees."

Same model, same question. The only change is that something outside it was allowed to run.

The rest of this lecture is about **who** decides to run it, and **where** the thing that runs lives.

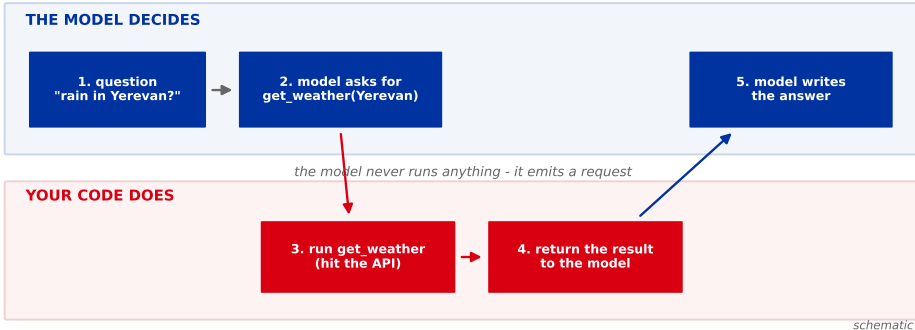
Outline

The one idea

Everything else in this lecture is a variation on it.

The model decides, your code does

Every approach in this lecture is a variation on these five steps



The model never runs anything. It emits a structured request - a piece of text saying which function it wants and with which arguments. Your program decides whether to honour it, runs it, and hands back the result.

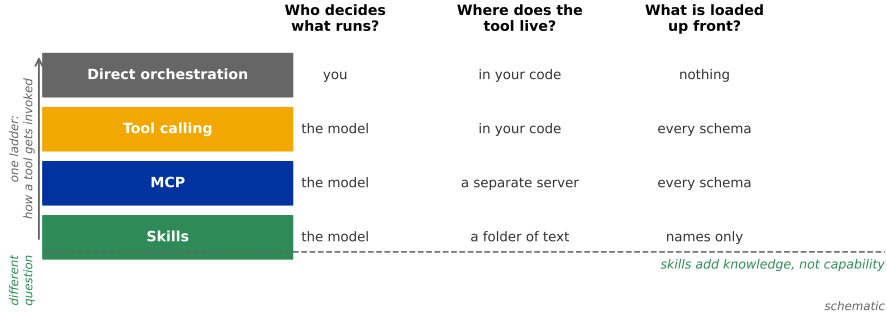
Why that split matters more than it looks

Because it means every safety and correctness question has an obvious home:

Question	**Where it is answered**
Can this user delete the database?	your code, before executing
Is this argument even valid?	your code, before executing
Should the model see this result?	your code, after executing
Which tool is right for this task?	the model

A model asking to run `delete_all_records()` is not a security breach. It is a **string**. It becomes a breach only if your code executes it without checking.

Four approaches, three questions



The first three rows are one ladder: each keeps the idea above it and changes exactly one thing about *how a tool gets invoked*.

Skills is not the fourth rung. It sits below the line because it answers a different question: the first three are ways to give the model a *capability*, skills is a way to give it *knowledge*. It shares the third column with them, and nothing else.

Direct orchestration

You decide what runs. The model only writes the sentence at the end.

The idea

You look at the question, you decide it needs the weather, you fetch it, and you paste the result into the prompt yourself.

```
weather = get_weather("Yerevan")  # you decided this
prompt = f"Weather: {weather}\n\nQuestion: {q}"
answer = model(prompt)
```

This is exactly what the RAG chapter did. Retrieval is direct orchestration where the tool happens to be a search index.

It is the simplest thing that works, it is completely predictable, and for a single known task it is often the right answer. People skip past it too quickly.

Where it stops

It works because *you* knew the question needed the weather.

Now the user asks:

> *"Should I take an umbrella to the meeting?"*

Now you need the weather *and* the calendar, and only for some phrasings of the question. You are now writing an if-statement over natural language.

Direct orchestration makes **you** the router. That is fine for one tool and hopeless for twenty, because routing natural language is the one job the model is better at than your code.

Tool calling

Hand the routing decision to the model, and keep the execution.

The idea

You describe your tools once, in a schema. The model reads the descriptions and, when it wants one, says so.

```
tools = [{"type": "function", "function": {
"name": "get_weather",
"description": "Current weather and today's forecast for a city.",
"parameters": {"type": "object",
"properties": {
"city": {"type": "string", "description": "e.g. Yerevan"},
"units": {"type": "string", "enum": ["metric", "imperial"]}},
"required": ["city"]}}}]
```

Note `enum`: it constrains what the model may put there. Anything JSON Schema can express - enums, arrays, nested objects, optional versus required - you can use.

The **description** is not documentation, it is the routing instruction. It is the only thing the model uses to decide whether this tool is the right one. A vague description is a routing bug.

What comes back is not an answer

When the model wants a tool, it does not reply with text. It replies with a request:

```
{"tool_calls": [{
"id": "call_01",
"function": {"name": "get_weather",
"arguments": "{\"city\": \"Yerevan\"}"}}]}
```

Your code runs it, then sends the result back as a new message, and the model produces the final sentence.

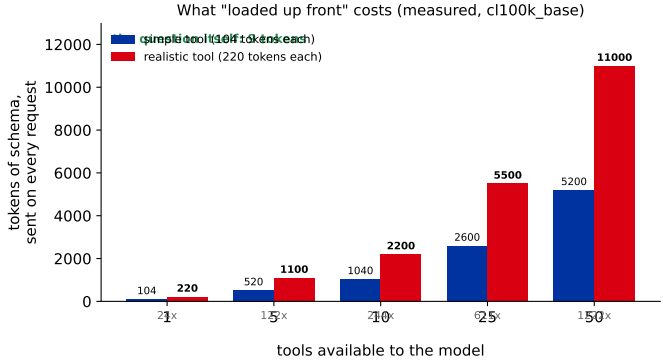
Two round trips to the model for one user question. This is the shape you will meet in every provider's API, under slightly different names.

Gotchas that will cost you an afternoon

- **Arguments come back as a JSON string**, not an object. It is generated text, so it can be malformed. Parse defensively.
- **The model can invent arguments** that are the right type and the wrong value. A valid city name is not a real city.
- **It can call the same tool twice**, or call nothing and answer anyway.
- **Your tool result must be sent back with the matching call id**, or the model loses track of which answer belongs to which request.

Every one of these is your code's problem, not the model's. The split from the first section is doing real work here: the model *decided*, so everything about whether that decision was safe or sane lands on your side of the line.

The cost nobody mentions



A one-parameter tool is **104 tokens**; a realistic one with a date range and filters is **220**. The question they serve is **9**. Schemas are sent on **every request**, used or not - so fifty tools is five to eleven thousand tokens per turn describing capabilities nobody asked for, and the model is choosing from fifty descriptions instead of five.

MCP

Same decision, different address. The tool moves out of your process.

The idea

With tool calling, the function lives in your codebase. You wrote it, you ship it, you own it.

Model Context Protocol (MCP) moves the tool behind a standard interface. A tool is a *server*; your application is a *client* that asks what it offers.

Tool calling	**MCP**
the function is in your code	the function is behind a protocol
you wrote every tool	you can plug in tools you did not write
adding a tool means a deploy	adding a tool means a config line

The model side does not change at all. It still just says "call this". MCP is an answer to **where the tool lives**, not to who decides.

Why a protocol is worth the ceremony

Because the interesting tools are the ones you did not write.

A filesystem server, a database server, a browser, someone's internal ticketing system - each written once, by whoever knows that system best, and usable by any client that speaks the protocol.

This is the same argument as any protocol in computing. `HTTP` did not make requests possible; it made it unnecessary for every client and every server to agree bilaterally in advance.

The cost is a process boundary: a server to run, to authenticate, to keep alive, and to blame when it is down. For one tool in one app, that ceremony is not worth it.

Skills

What if the thing you are adding is not code, but knowledge?

The idea

Not everything you want to add is executable. Often what the model is missing is not a capability but an **instruction**: your house style, your escalation policy, the six steps your team follows to close a ticket.

A **skill** is a folder with a markdown file in it. The model is told the skill's name and one-line description, and reads the body only when it decides the skill applies.

```
---
name:  close-ticket
description:  Use when a support ticket needs closing.
---
1.  Check the customer replied.
2.  Summarise the resolution in two sentences.
3.  Tag the owning team.  Never close without a tag.
```

Why this is not just a longer prompt

Because of the third column from the ladder: **what is loaded up front**.

	Loaded always	**Loaded on demand**
System prompt	everything	nothing
Tool schemas	every schema, every request	nothing
Skills	**name + one line**	**the whole body**

You can have two hundred skills and pay for two hundred single lines. That is the entire trick, and it is the direct answer to the token cost measured two sections ago.

The price: the model must choose correctly from one-line descriptions alone. Same failure mode as a bad tool description, one level up.

The loop

Everything so far ran once. An agent is what happens when it runs again.

One call is not an agent

Every approach so far had the same shape: the model asks for one thing, you run it, it answers. One round trip, then done.

Now ask something that cannot be answered in one:

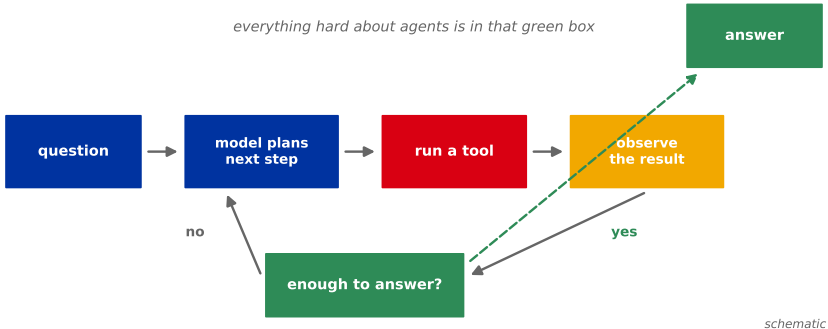
> *"Find the cheapest flight to Tbilisi next weekend and check if I am free."*

You cannot know how many tool calls that takes. It depends on what the first one returns.

So stop fixing the number. Let the model keep going until it decides it is done. That decision is the whole difference between *tool use* and an *agent*.

The loop

An agent is the same five steps, with a decision that sends you back



Structurally it is a `while` loop with one exit.

The loop, written out

```
while True:
reply = model(messages, tools=tools)
if not reply.tool_calls:    # no request = it is done
return reply.text
for call in reply.tool_calls:
result = run(call)      # your code, your checks
messages.append(result)
```

That really is the whole mechanism. **And it is not what you should ship**, because every guard from the last two slides is missing from it.

Add the turn cap, the repeat detector, the spend cap, argument validation and the transcript log, and this doubles in size. The *idea* stays small; the *production loop* never does. Be suspicious of any framework that shows you only the seven lines.

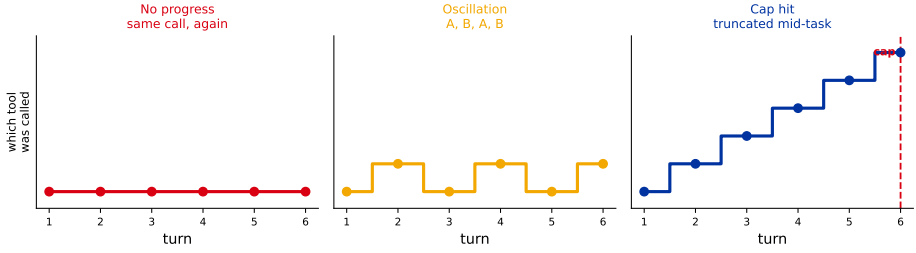
The green box is where the difficulty lives

"Enough to answer?" is not a function you write. In practice it is the model itself: on each turn it either requests another tool, or stops requesting and writes prose. The absence of a tool call *is* the stopping signal.

Which means **nothing guarantees it stops.** There is no proof of termination, no bound on the number of turns, and no error raised when it goes wrong. A loop that never converges looks exactly like a loop that is still working.

So you impose a cap. And the moment you do, you have created a second failure mode: a task that was going to succeed on turn twelve, cut off at turn ten.

Three ways it fails to stop



No progress: the same call with the same arguments, because the result did not change the model's mind. **Oscillation**: two tools that undo each other. **Cap hit**: real work, truncated - the only one of the three where the loop was doing fine.

What to actually do about it

- **Cap the turns**, and treat hitting the cap as a *failure to report*, not a silent truncation. The user must be told the task was cut off.
- **Detect repeats**: same tool, same arguments, twice in a row is a loop, not progress. Break and say so.
- **Cap the spend**, not just the turns. Turns are a proxy; tokens are the bill.
- **Log every turn**: the tool name, the exact arguments, the raw result, the token count and the elapsed time. When an agent does something strange, that transcript is the only artefact that explains it - and it is the first thing people forget to keep.

None of these makes the loop correct. They make it **bounded and legible**, which is what you can actually have.

Choosing

They are layers, not competitors.

Decision guide

If...	Reach for
one known task, fixed shape	**direct orchestration** - do not over-engineer
the user's phrasing decides what runs	**tool calling**
the tool is not yours, or is shared across apps	**MCP**
you are adding knowledge, not capability	**skills**
the number of steps is unknown in advance	**an agent loop**, with a cap

Start at the top and move down only when something forces you. Every row below the first buys flexibility with a failure mode, and the failure modes compound. They also **stack**: a production system is usually a loop, over tool calls, some arriving by MCP, guided by skills.

Recap

- **The model decides, your code does.** It never executes anything; it emits a request. Every safety check lives on your side of that line.
- **Direct orchestration** makes you the router. Simple, predictable, and it stops scaling the moment phrasing decides what runs.
- **Tool calling** hands routing to the model. The description is the routing instruction, and the schemas cost you tokens on every request - 104 for one tool, against a 9-token question.
- **MCP** answers *where the tool lives*, not who decides. It buys tools you did not write, and costs you a process boundary.
- **Skills** add knowledge rather than capability, and are loaded on demand - which is the direct answer to the schema token cost.
- **An agent** is the same machinery with a decision that sends you back. Nothing guarantees it stops, so bound it and log it.

One line separates a tool from an agent.

It is the line that decides whether to go around again - and it is the one nobody can prove terminates.